CSC 370: Databases ERD's to FD's?

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Intended Learning Outcome:

Given an entity-relationship diagram, you can:

Verify the set of relations and constraints that it implies

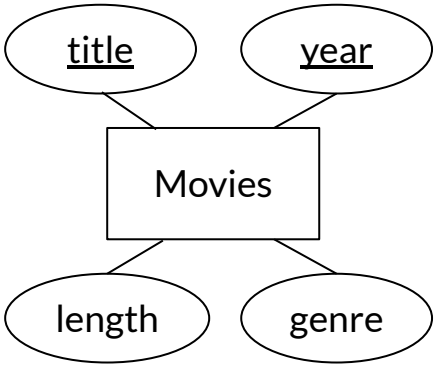
Outline

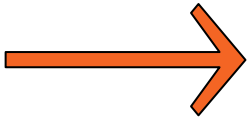
1. Can we trust the “rules” we learned for converting ERDs into relations?
2. Where are functional dependencies implied in an ERD?
3. How can we verify normality in the Entity/Relationship data model?

Revisiting the E/R - Relational Link...

Basic Conversion Rules: Strong Entity Sets

- Transform each entity set into a relation
  - All attributes appear once in the relation
  - The identifier becomes a key



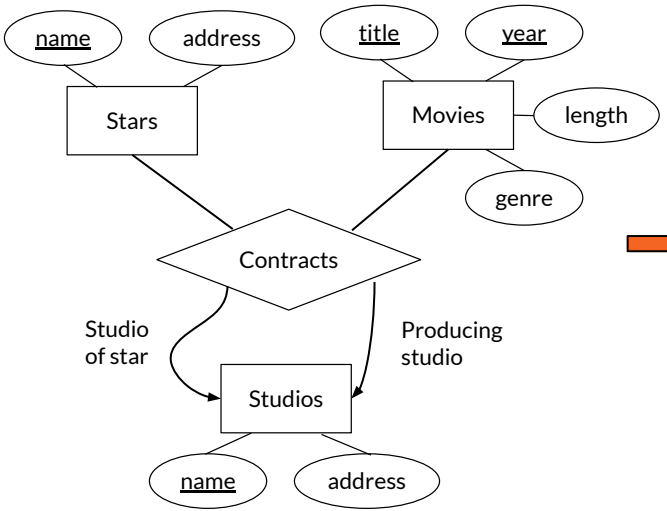


Movies(<u>title</u>, <u>year</u>, length, genre)

Source: Garcia-Molina et al. (2007), §4.5

Basic Conversion Rules: Relationships

- Transform each relationship into a relation
  - For each entity set involved in the relationship, its identifiers become attributes
  - Each attribute on the relationship also becomes an attribute of the relationship
  - The combination of “foreign” identifiers becomes a key

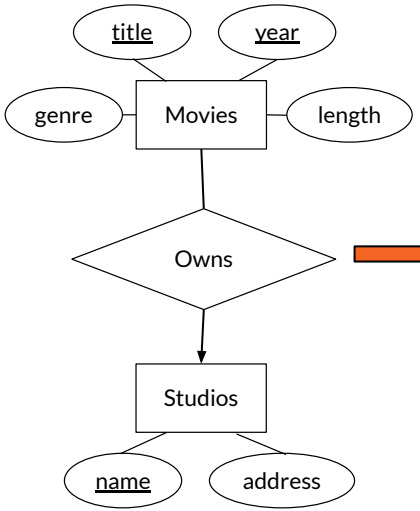


Contracts(starName, title, year, studioOfStar, producingStudio)

Source: Garcia-Molina et al. (2007), §4.5

Basic Conversion Rules: Combining Relat'ns

- We can combine relations from an entity set *E* and a many-one relationship *R* to *F*
  - The union of their attributes becomes the new schema
  - In the end, this will include all attributes of *E*, all attributes of *R*, and the identifier of *F*



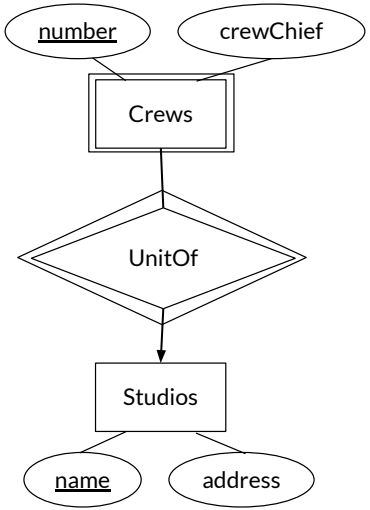
Movies(<u>title</u>, <u>year</u>, genre, length)
Studios(<u>name</u>, address)
Owns(<u>title</u>, <u>year</u>, <u>studioName</u>)

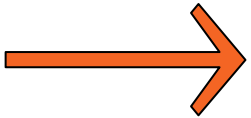
Movies(<u>title</u>, <u>year</u>, genre, length, studioName)
Studios(<u>name</u>, address)

Source: Garcia-Molina et al. (2007), §4.5

Basic Conversion Rules: Weak Entity Sets

- A weak entity set *E* and a many-one relationship *R* to *F* are handled similarly to many-to-one
  - The attributes of the relation are all those of *E* and *R* and the identifier attributes for *F*
  - A key for *E* becomes the union of identifier attributes in *E* and in *F*



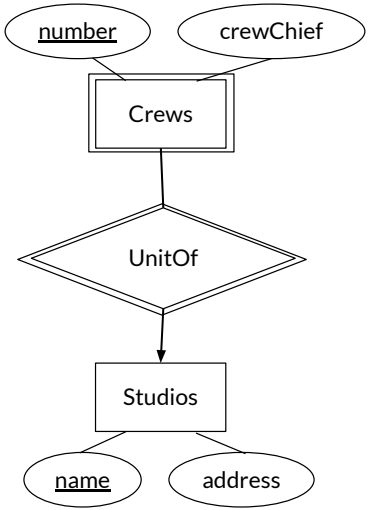


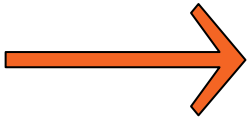
Studios(<u>name</u>, address)
Crews(<u>studioName</u>, <u>number</u>, crewChief)

Source: Garcia-Molina et al. (2007), §4.5

Basic Conversion Rules: Weak Entity Sets

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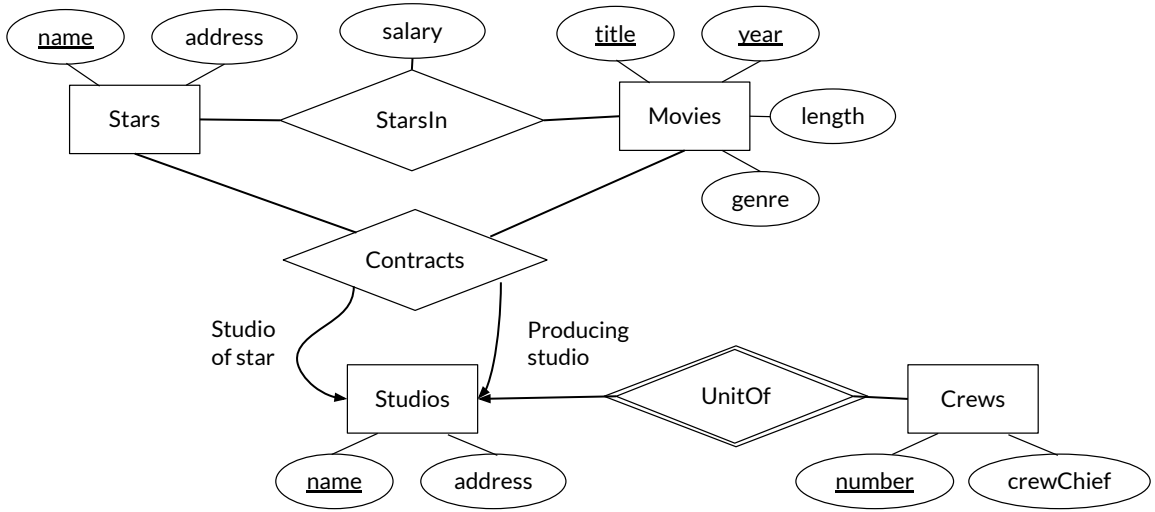




Studios(<u>name</u>, address)
Crews(<u>studioName</u>, <u>number</u>, crewChief)

Source: Garcia-Molina et al. (2007), §4.5

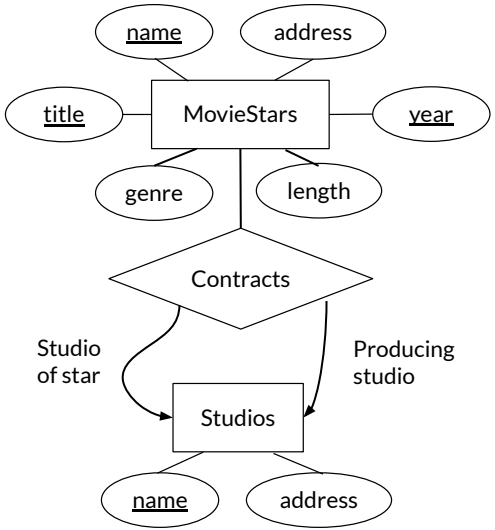
How many relations are needed?



❑ 1 ❑ 2 ❑ 3 ❑ 4 ❑ 5 ❑ 6 ❑ 7

Sanity Checking our Rules

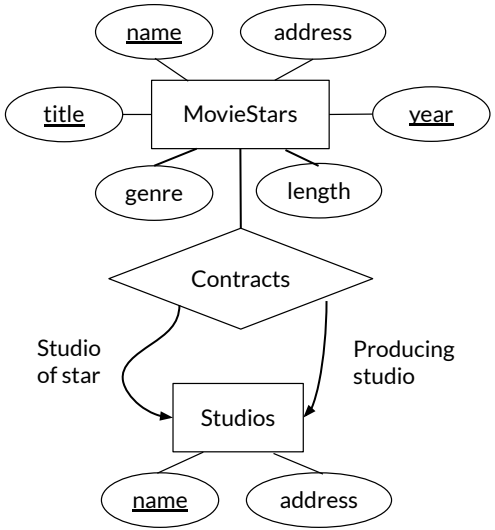
- What happens in this example?



MovieStars(starName, title, year, address, genre, length)
Contracts(starName, title, year, studioOfStar, producingStudio)

Sanity Checking our Rules

- What happens in this example?
  - Now we have claimed that starName $\not\rightarrow$ address!
  - For this conversion to work, the ERD had to already be normalised!

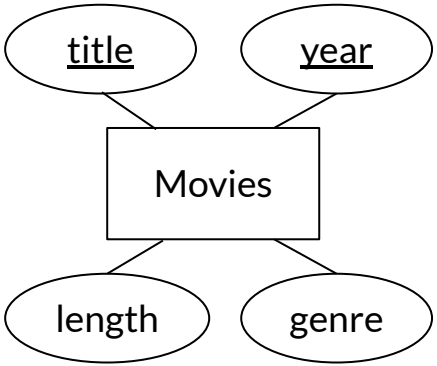


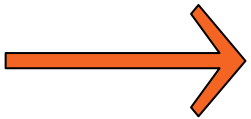
MovieStars(<u>starName</u>, <u>title</u>, <u>year</u>, address, genre, length)
Contracts(<u>starName</u>, <u>title</u>, <u>year</u>, <u>studioOfStar</u>, <u>producingStudio</u>)

Reading FDs from an ERD...

FD Rules: Strong Entity Sets

- The identifier of a strong entity set functionally determines all non-identifier attributes

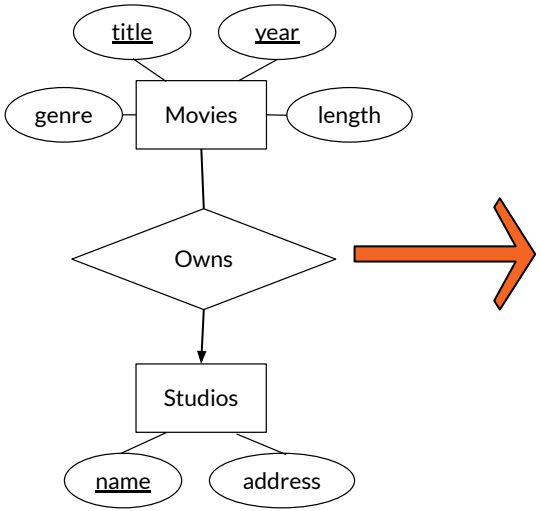




title year → length genre

FD Rules: Many-to-One Relationships

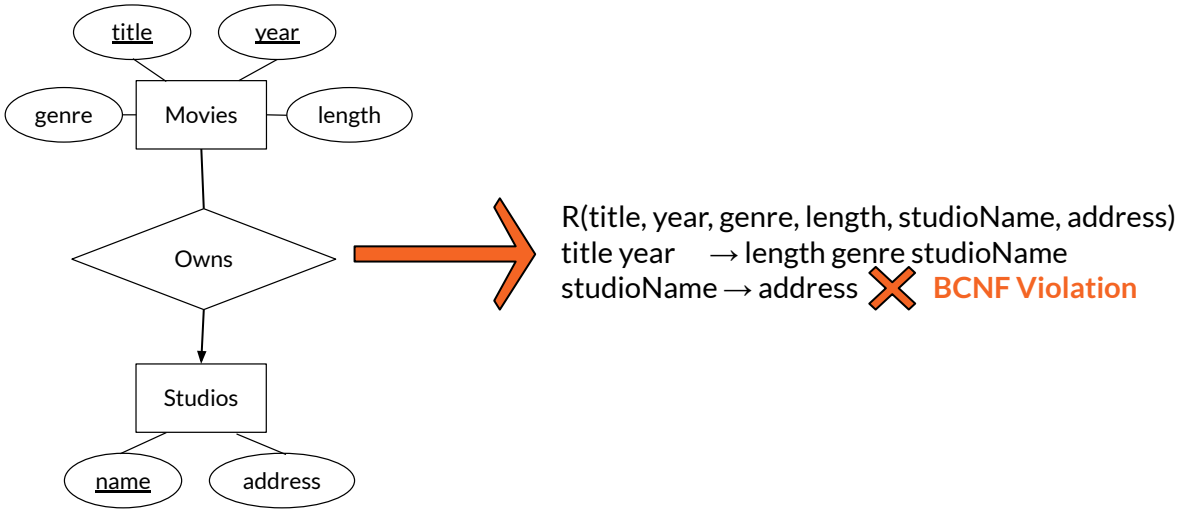
- If an entity set *E* has a many-one relationship *R* to *F:*
  - Then if two tuples in R contain the same identifier of E, they must also have the same identifier of F



R(title, year, genre, length, studioName, address)
title year → length genre
studioName → address

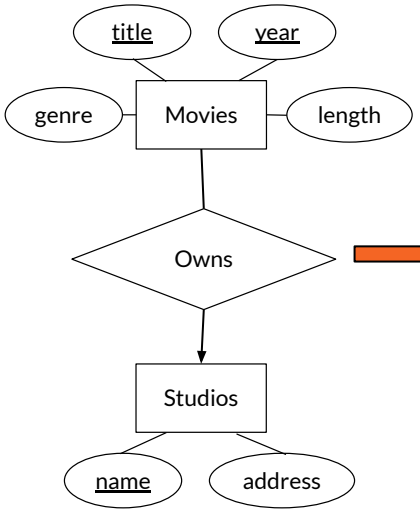
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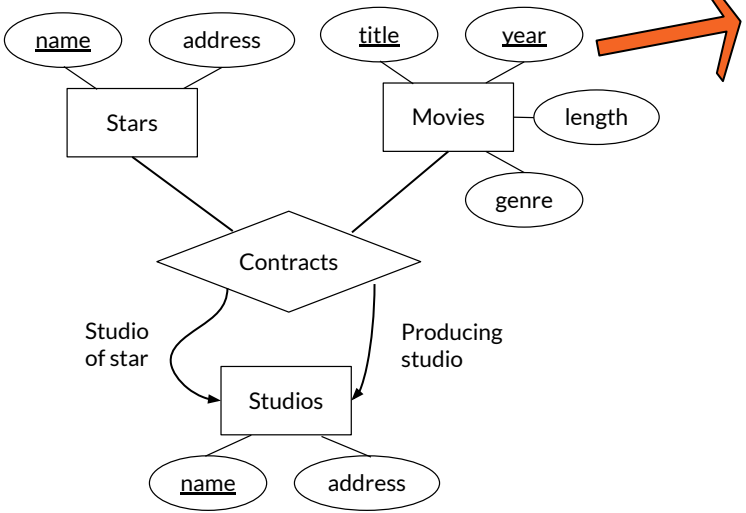


R(title, year, genre, length, studioName, address)
title year → length genre studioName
studioName → address **BCNF Violation**

Movies(<u>title</u>, <u>year</u>, genre, length, studioName)
Studios(<u>name</u>, address)

FD Rules: Many-to-Many Relationships

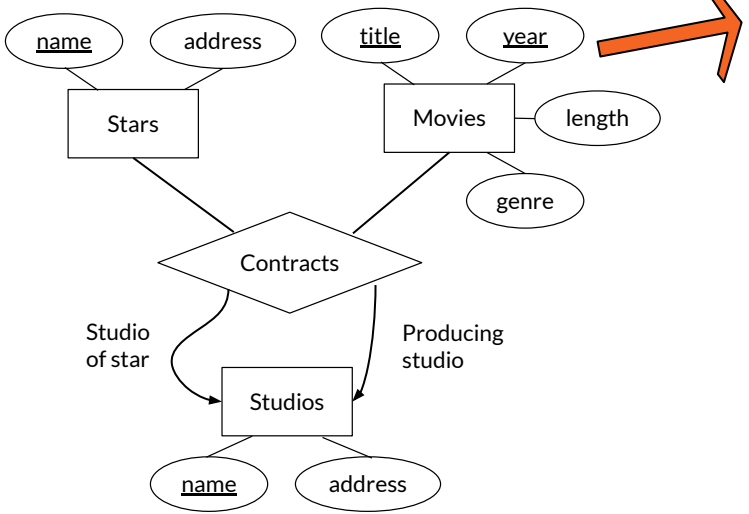
- Both entity sets end up producing BCNF violations



R(starName, starAddress, title, year, length, genre, studioOfStar,producingStudio)
title year → length genre
starName → starAddress
starName title year → studioOfStar producingStudio

FD Rules: Many-to-Many Relationships

- Both entity sets end up producing BCNF violations

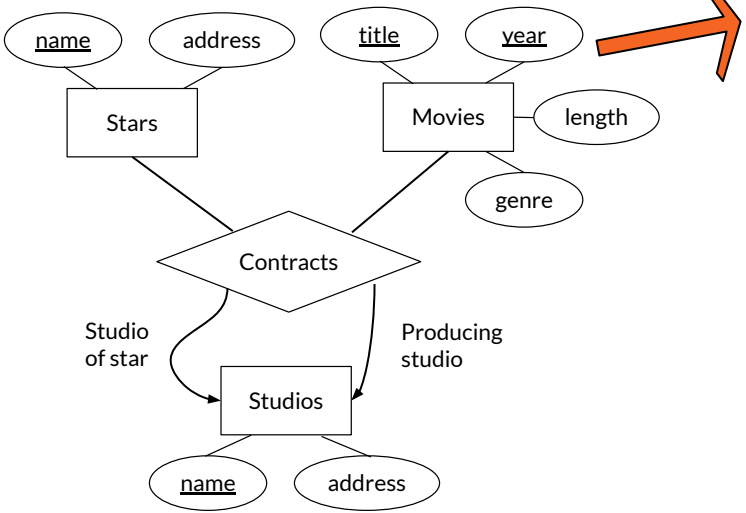


R(starName, starAddress, title, year, length, genre, studioOfStar,producingStudio)
title year → length genre
starName → starAddress
starName title year → studioOfStar producingStudio

BCNF Violation

FD Rules: Many-to-Many Relationships

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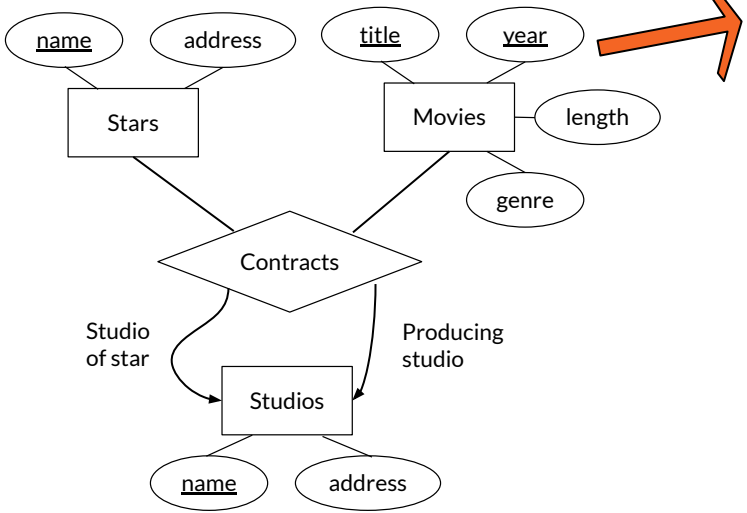
R(starName, starAddress, title, year, length, genre, studioOfStar,producingStudio)
title year → length genre
starName → starAddress
starName title year → studioOfStar producingStudio

BCNF Violation

Stars(starName, starAddress)
R(starName, title, year, length, genre, studioOfStar,producingStudio)
title year → length genre
starName title year → studioOfStar producingStudio

FD Rules: Many-to-Many Relationships

- Both entity sets end up producing BCNF violations

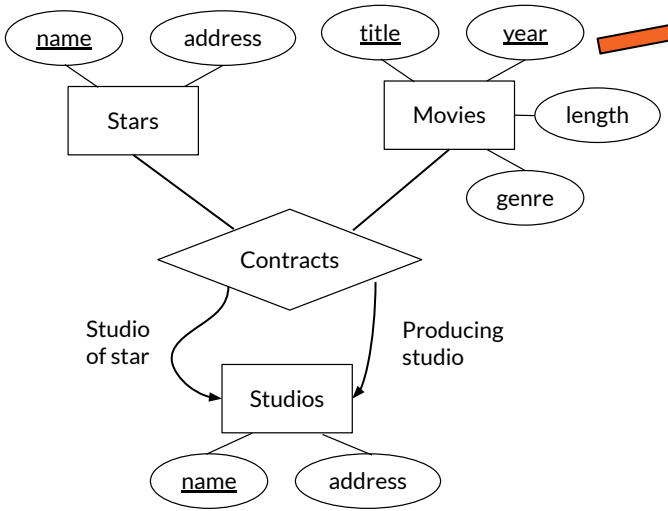


R(starName, starAddress, title, year, length, genre, studioOfStar,producingStudio)
title year → length genre
starName → starAddress **BCNF Violation**
starName title year → studioOfStar producingStudio

Stars(starName, starAddress)
R(starName, title, year, length, genre, studioOfStar,producingStudio)
title year → length genre **BCNF Violation**
starName title year → studioOfStar producingStudio

FD Rules: Many-to-Many Relationships

- Both entity sets end up producing BCNF violations



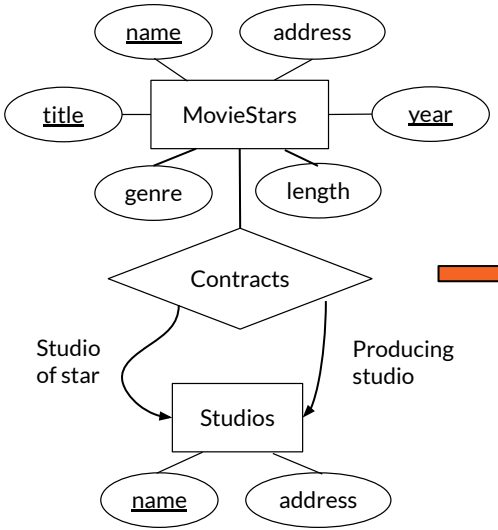
R(starName, starAddress, title, year, length, genre, studioOfStar,producingStudio)
title year → length genre
starName → starAddress **BCNF Violation**
starName title year → studioOfStar producingStudio

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starName title year → studioOfStar producingStudio

Sanity Checking our FD Rules

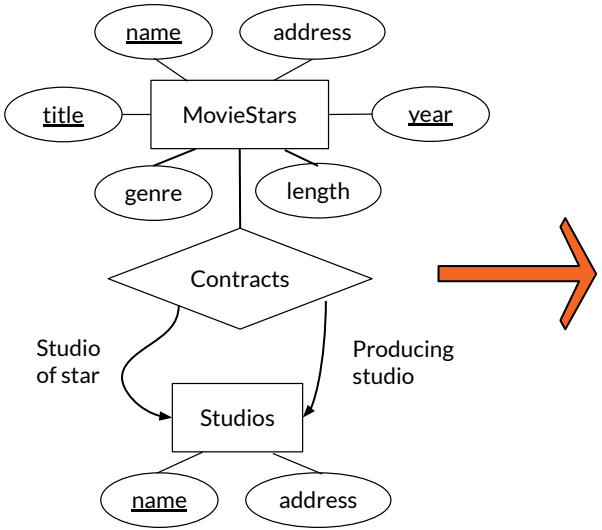
- What happens in this example when we consider FDs?



R(starName, title, year, starAddress, genre, length, studioOfStar, producingStudio, starStudioAddress, producingStudioAddress)
starName title year → starAddress genre length studioOfStar producingStudio
studioOfStar → starStudioAddress
producingStudio → producingStudioAddress

Sanity Checking our FD Rules

- What happens in this example when we consider FDs?



R(starName, title, year, starAddress, genre, length, studioOfStar, producingStudio, starStudioAddress, producingStudioAddress)
starName title year → starAddress genre length studioOfStar producingStudio
studioOfStar → starStudioAddress
producingStudio → producingStudioAddress

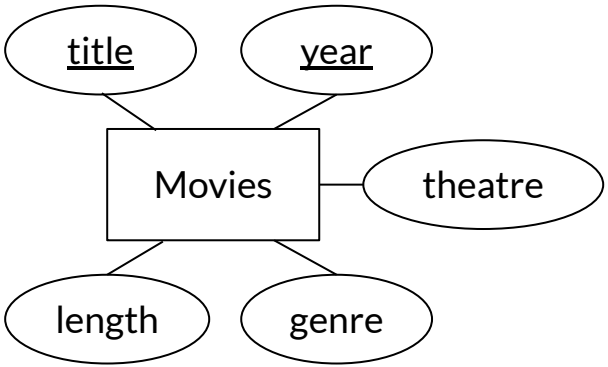
Our rules don't reveal:
starName → starAddress
title year → genre length

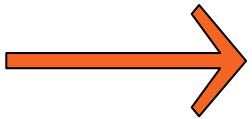
So of course normalisation would fail!

Verifying normality...

Verifying Normality: Strong Entity Sets

- Are all non-identifier attributes functionally determined by the identifier?
  - If not, this entity set needs to be split up

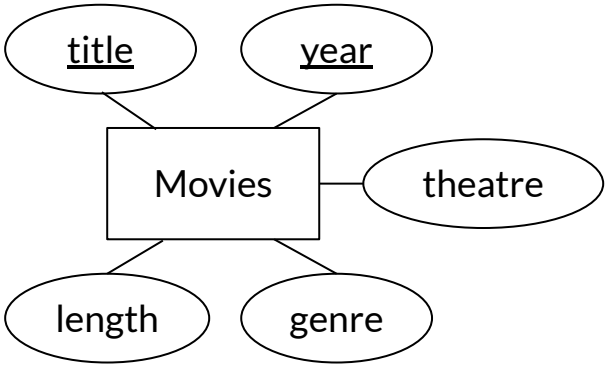




title year → length genre

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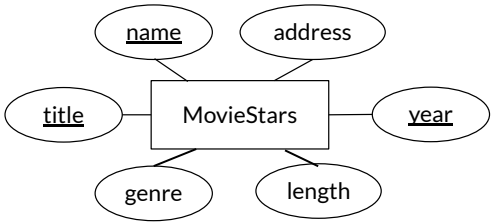


title year → length genre

✗ Identifier is not a key!

Verifying Normality: Strong Entity Sets

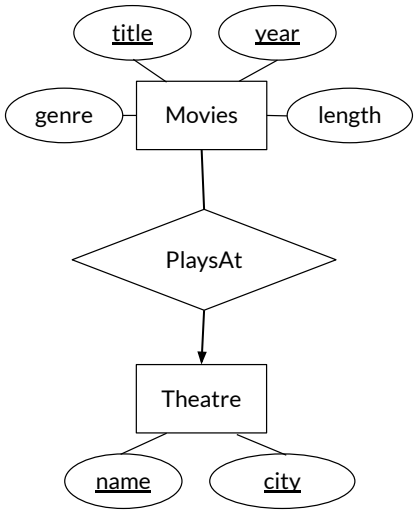
- Are there other functional dependencies with a proper subset of the identifier?
  - If so, this entity set needs to be split up



Missing:
title year → length genre
starName → starAddress

Verifying Normality: Relationships

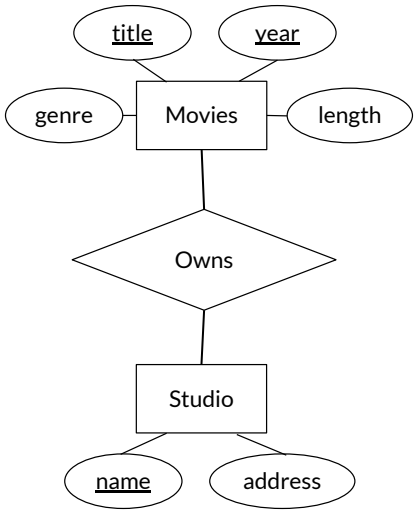
- For a many-to-one relationship, does the “many” identifier functionally determine the “one” identifier?
  - If not, then the multiplicity is wrong



Incorrect implication:
title year → theatreName

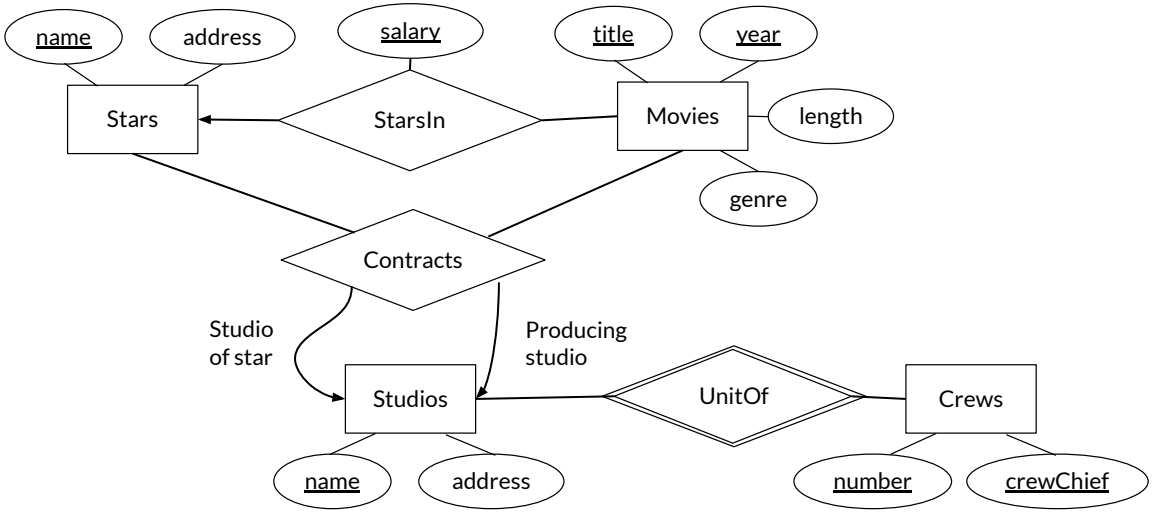
Verifying Normality: Relationships

- For a many-to-many relationship, are there no functional dependencies among identifiers of the participating entity sets?
  - If so, then the multiplicity is wrong



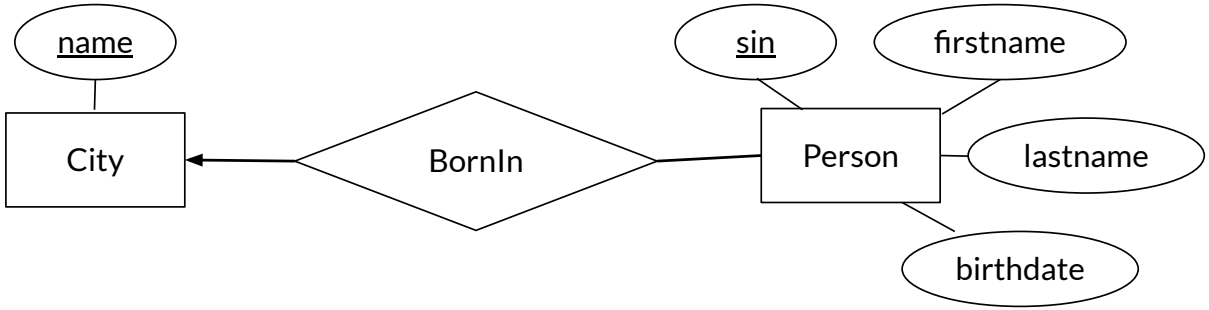
Missing functional dependency:
title year → studioName

Which FDs are incorrect below?



Time Allotted: 120 seconds

What are the BCNF relations?



Time Allotted: 120 seconds